Assignment 2: Word Embeddings Comparison, Visualization, and Evaluation

Natural Language Processing - Fall 2026

Objective

In this assignment, you will preprocess a text dataset, generate and compare Word2Vec Skipgram and GloVe embeddings, and visualize them. Additionally, you'll evaluate the embeddings to determine which model best captures the semantic relationships within the dataset.

Instructions

Dataset

Document Classification Dataset: containing documents in five different domains: `politics`, `sport`, `tech`, `entertainment` and `business`.
Dataset Link: https://www.kaggle.com/datasets/sunilthite/text-document-classification-dataset

Preprocessing

Clean the text data by:

- Removing punctuation, stop words, and special characters.
- Lowercasing all text.
- Tokenizing sentences into words.
- Lemmatizing or stemming to normalize the vocabulary.

Word Embedding Models

Implement or load the following word embeddings:

- **Word2Vec Skipgram:** Uses context to predict a target word.
- **GloVe:** Uses word co-occurrence statistics to capture global word relationships..

Comparative Analysis

Document the differences between embeddings generated by Skipgram and GloVe. Show differences in word relationships or similarities among the embeddings.

Visualization

Use **PCA** and **t-SNE** to reduce embedding dimensions and create 2D visualizations and plot the embeddings to observe clustering and spatial relationships. Explain any patterns observed in each model's clusters and the difference between each visualization method.

Evaluation

Assess each embedding method based on:

- **Semantic Similarity:** How well each model captures similar words or related concepts (e.g., are synonyms close together?).
- **Topic Coherence:** Are words from the same topic or context (e.g., technology, emotions) grouped together?
- **Performance on Downstream Tasks:** use a classification model and compare accuracy across embeddings

Document your findings, highlighting which model best captures the relationships within the dataset. Discuss any limitations or observations.

Submission Requirements

1. A Jupyter notebook with code, outputs, and visualizations, which runs top to bottom without errors.
2. A report (PDF) summarizing your methodology and key results.
3. Comment your code and explain your reasoning in markdown cells.
4. Plagiarism will be heavily penalized

Deadline:

Wednesday 28/10/2026 at 11:59 pm
Form Link: https://forms.gle/bx7zua9uLLZBixHP9